PROJECT ATLAS

URS-001: USER REQUIREMENTS SPECIFICATION

Internal Codename: Atlas | Product: Master Flightline Clock

1. Document Control

Document ID	URS-001
Version	1.0
Status	APPROVED
Approved By	Peter Lochhead Hepburn (Product Owner)
Issue Date	2026-06-30

1.1 Revision History

Ver	Date	Author	Description
0.1	2026-06-30	Gemini (Architect)	Initial draft following reset.
1.0	2026-06-30	Peter Hepburn	Formal Baseline for Phase 0.

2. Introduction & Purpose

The purpose of this User Requirements Specification (URS) is to define the critical requirements for the Master Flightline Clock (Project Atlas). This document aligns with GAMP 5 principles by focusing on "Intended Use" and providing the baseline against which the system will eventually be validated during Operational (OQ) and Performance Qualification (PQ).

3. User Requirements

3.1 Operational Controls (Traceability ID: UR-OP)

ID	Requirement Description	Priority
UR-OP-01	The system shall utilize a "Concurrent State Model" allowing the operator to configure a **Next Heat** while an **Active Heat** is being timed and displayed.	Must
UR-OP-02	The transition from State_Next to State_Active shall be triggered by a physical or software "Activate" command.	Must
UR-OP-03	Activation of a new heat shall be strictly prohibited if a current heat is already in progress.	Must

3.2 Display & Timing (Traceability ID: UR-DT)

ID	Requirement Description	Priority
UR-DT-01	The system shall show local PC time in HH:MM:SS format at all times.	Must
UR-DT-02	The system shall perform a staged countdown: First, the "Insertion Window" (Green), followed immediately by the "Heat Duration" (Red).	Must
UR-DT-03	Timing calculations must be deterministic, syncing with hardware interrupts to prevent drift over 60-minute durations.	Must

3.3 Performance & Safety (Traceability ID: UR-PS)

ID	Requirement Description	Priority
UR-PS-01	A "Master Cancel" function shall be available to arrest all timers and zero the display in the event of a race incident.	Must
UR-PS-02	The system shall maintain state and configuration data locally to survive power cycles without data loss.	Should

Product Owner Approval:

I confirm that the requirements listed in this document accurately reflect the operational needs of the Southern Soaring League competitions.

Signed: ________________________________ Date: _____________